

DASF-Net: Dual-Adversarial Spectral Fusion for Cross-Lingual and Adversarially Robust Audio Deepfake Detection

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Abstract—Audio deepfake detectors are trained and evaluated almost exclusively on high-resource languages and on a narrow family of neural vocoders. When such a detector meets a new language synthesised by a structurally different generator, two failure modes compound: the learned features encode language- and corpus-specific cues rather than synthesis artifacts, and the decision boundary remains trivially breakable by gradient-based perturbation. We propose DASF-Net, a detection architecture built around the observation that these two failures are both *adversarial* problems and can be attacked with a single training formulation. DASF-Net couples (i) a dual-branch spectral front-end that presents the same utterance as a sparse cepstral view (LFCC) and a dense spectral view (log-Mel) to one Siamese-shared, ImageNet-pretrained Xception trunk; (ii) a *Cross-Attention Spectral Fusion* (CASF) module that lets the two views query one another and combines them through a learned gate, so the network selects per utterance which representation carries the artifact; and (iii) a *dual-adversarial* objective in which a gradient-reversed language discriminator drives the embedding toward language invariance while adaptive FGSM/PGD adversarial training drives it toward perturbation robustness. A three-stage curriculum trains the parts in a stable order. We instantiate the design on the English WaveFake corpus (117,985 utterances, seven GAN- and flow-based vocoders) as source domain and the Bengali BanglaFake corpus (25,520 utterances, one VITS end-to-end system) as target domain, a pairing that varies language and synthesis paradigm at once. We further specify a Confound-Controlled Evaluation Protocol that neutralises the speaker, gender and class-prior imbalances which otherwise inflate Bengali detection scores. This paper presents the architecture, the training algorithms and the experimental design; the empirical study is reported separately.

Index Terms—audio deepfake detection, cross-lingual generalisation, adversarial robustness, domain-adversarial training, cross-attention fusion, Mel-spectrogram, Xception, Bengali speech, low-resource language

I. INTRODUCTION

Neural vocoders and end-to-end text-to-speech (TTS) systems now produce speech that listeners routinely fail to separate from genuine recordings. The resulting misuse — voice-impersonation fraud, fabricated evidence, political disinformation — has driven a decade of work on automatic detection. That work is overwhelmingly English, and overwhelmingly built on a single family of generators. Bengali, with more than 230 million native speakers, has almost no detection literature: BanglaFake first made Bengali deepfake audio available at

scale but trained no detector, and “Zero-Shot to Zero-Lies” first benchmarked detectors on it, reporting a best zero-shot result of 53.80% accuracy (46.20% EER, Wav2Vec2-XLSR-53) and a best fine-tuned result of 79.17% accuracy (24.35% EER, ResNet18).

Those numbers expose a gap that is not merely quantitative. A detector that transfers this poorly is not simply under-trained; it has learned the wrong thing. Its features encode which corpus, which speaker and which language produced an utterance, rather than what a synthesiser did to it. The same detector is also, as a large body of adversarial work shows, breakable by perturbations far below the audible threshold: Kawa *et al.* degrade an LCNN countermeasure from 0.022 to 0.785 EER under FGSM, and our own prior study reproduces the pattern on a much larger ImageNet-pretrained Xception detector.

A. From measuring a gap to closing it

Prior cross-lingual studies characterise the degradation and stop there. Our position is that the two failures above share a structure. In both cases the network has latched onto a nuisance direction in feature space — language and corpus identity in one case, a locally exploitable gradient direction in the other — and in both cases the standard remedy is to introduce an adversary that penalises exactly that direction. Domain-adversarial training supplies the first adversary; FGSM/PGD adversarial training supplies the second. Placing both on a single shared embedding is, to our knowledge, new for audio deepfake detection, and it turns the cross-lingual question from a measurement exercise into an architectural one.

A second, independent question concerns *which* spectral representation a convolutional detector should consume. Linear Frequency Cepstral Coefficients (LFCC) have been the default for spoofing countermeasures since the Gaussian-Mixture-Model era, yet our prior study showed that for CNNs the ordering reverses: dense 128-bin log-Mel spectrograms outperform 20-coefficient LFCC by a consistent 33–35% relative EER margin across all seven WaveFake vocoders, because dense two-dimensional inputs suit both convolutional inductive bias and ImageNet transfer. That study treated the two representations as mutually exclusive alternatives and

explicitly left hybrid representations as future work. DASF-Net takes up that thread: rather than choosing, it consumes both and learns a per-utterance gate between them.

B. Contributions

- 1) **DASF-Net**, an architecture that presents sparse cepstral and dense spectral views of one utterance to a single Siamese-shared Xception trunk and fuses them with a two-way cross-attention module and a learned gate (Section IV).
- 2) A **dual-adversarial objective** that combines a gradient-reversed language discriminator with adaptive FGSM/PGD adversarial training on the same fused embedding, together with a three-stage curriculum that makes the combination trainable (Section V).
- 3) A **Confound-Controlled Evaluation Protocol** (CCEP) for Bengali–English deepfake transfer, which neutralises the speaker, gender and class-prior artifacts that make naive BanglaFake accuracy figures uninterpretable, and a corrected, unit-consistent adversarial threat model (Sections IV-G and VI).
- 4) An experimental design that separates the contribution of each component through ablation, so that any cross-lingual gain can be attributed rather than merely observed (Section VI).

This paper is deliberately scoped to design and methodology. No empirical results are reported here.

II. RELATED WORK

A. Datasets and benchmarks

ASVspoof and the ADD challenge series standardised English and Chinese anti-spoofing evaluation. WaveFake contributed 117,985 utterances spanning seven neural vocoders derived from LJSpeech, enabling leave-one-vocoder-out study of cross-generator generalisation. BanglaFake pairs 12,260 real Bengali utterances (SUST TTS Corpus, Mozilla Common Voice) with 13,260 utterances from a VITS system trained on the SUST corpus; its authors assess dataset quality by Mean Opinion Score and a t-SNE view of MFCC features, reporting substantial overlap between the real and synthetic clusters, but train no detector. The Zero-Shot to Zero-Lies benchmark supplied the first Bengali detection benchmark over zero-shot pretrained models (Wav2Vec2-XLSR-53, Whisper, PANNs-CNN14, WavLM, AST) and fine-tuned architectures (Wav2Vec2-Base, LCNN, LCNN-Attention, ResNet18, ViT-B/16, CNN-BiLSTM).

B. Feature representations for CNN detectors

LFCC with a GMM back-end long dominated countermeasure design, and the WaveFake study adopted it as its baseline. Our prior work showed the advantage is back-end-specific: a dense 128-bin log-Mel spectrogram transfers far better into an ImageNet-pretrained Xception CNN than a sparse 20-coefficient cepstral representation, because the former retains the two-dimensional time–frequency structure that convolution and ImageNet initialisation both assume. Whether that

advantage is specific to GAN-vocoder artifacts, or survives a change of synthesis paradigm and language, motivates the present architecture. Multi-view and multi-branch front-ends are established elsewhere in speech, but we are not aware of a learned, attention-gated fusion of cepstral and spectral views for deepfake detection.

C. Domain-adversarial and cross-lingual transfer

Domain-adversarial neural networks attach a domain classifier through a gradient reversal layer so that the shared encoder is trained to be uninformative about domain while remaining discriminative for the task. The technique is standard in vision and in speech recognition, but it has not, to our knowledge, been applied to make an audio deepfake detector language-invariant. Both the BanglaFake dataset paper and the Zero-Shot to Zero-Lies benchmark name cross-lingual generalisation as open for Bengali; neither proposes a mechanism for it.

D. Adversarial robustness of countermeasures

FGSM and PGD are the standard white-box stress tests. Kawa et al. demonstrated catastrophic LCNN degradation under FGSM and partial recovery through adversarial training at a clean-accuracy cost. Our prior study reproduced this on Xception with an adaptive scheme that samples attacks in proportion to a momentum-smoothed estimate of their recent success. No prior work examines adversarial robustness for Bengali deepfake detection, nor whether robustness acquired in one language survives transfer to another.

III. DATASETS

A. WaveFake (English, source domain)

WaveFake contains 117,985 utterances (13,100 real, 104,885 synthetic) built from the single-speaker LJSpeech corpus: MelGAN, MelGAN-Large, Parallel WaveGAN, Multi-Band MelGAN, Full-Band MelGAN, HiFi-GAN and WaveGlow, plus a full TTS pipeline. All are GAN-based or GAN-adjacent flow vocoders that *resynthesise* a waveform from acoustic features of real speech.

B. BanglaFake (Bengali, target domain)

BanglaFake contains 25,520 utterances: 12,260 real (10,000 SUST TTS Corpus, 2,260 Common Voice across five speakers) and 13,260 synthesised by a single VITS model trained from scratch on the SUST corpus. VITS is an end-to-end variational/flow architecture that generates speech from text without an intermediate acoustic-feature stage, and is therefore structurally unlike every WaveFake generator. Audio is distributed as 22,050 Hz WAV, 6–7 s per clip, in LJSpeech-style layout.

C. Why this pairing, and what it confounds

Because language and synthesis paradigm change together, English $\leftrightarrow$ Bengali transfer probes cross-lingual *and* cross-architecture generalisation simultaneously. That is the property we want, but the pairing also carries three confounds that

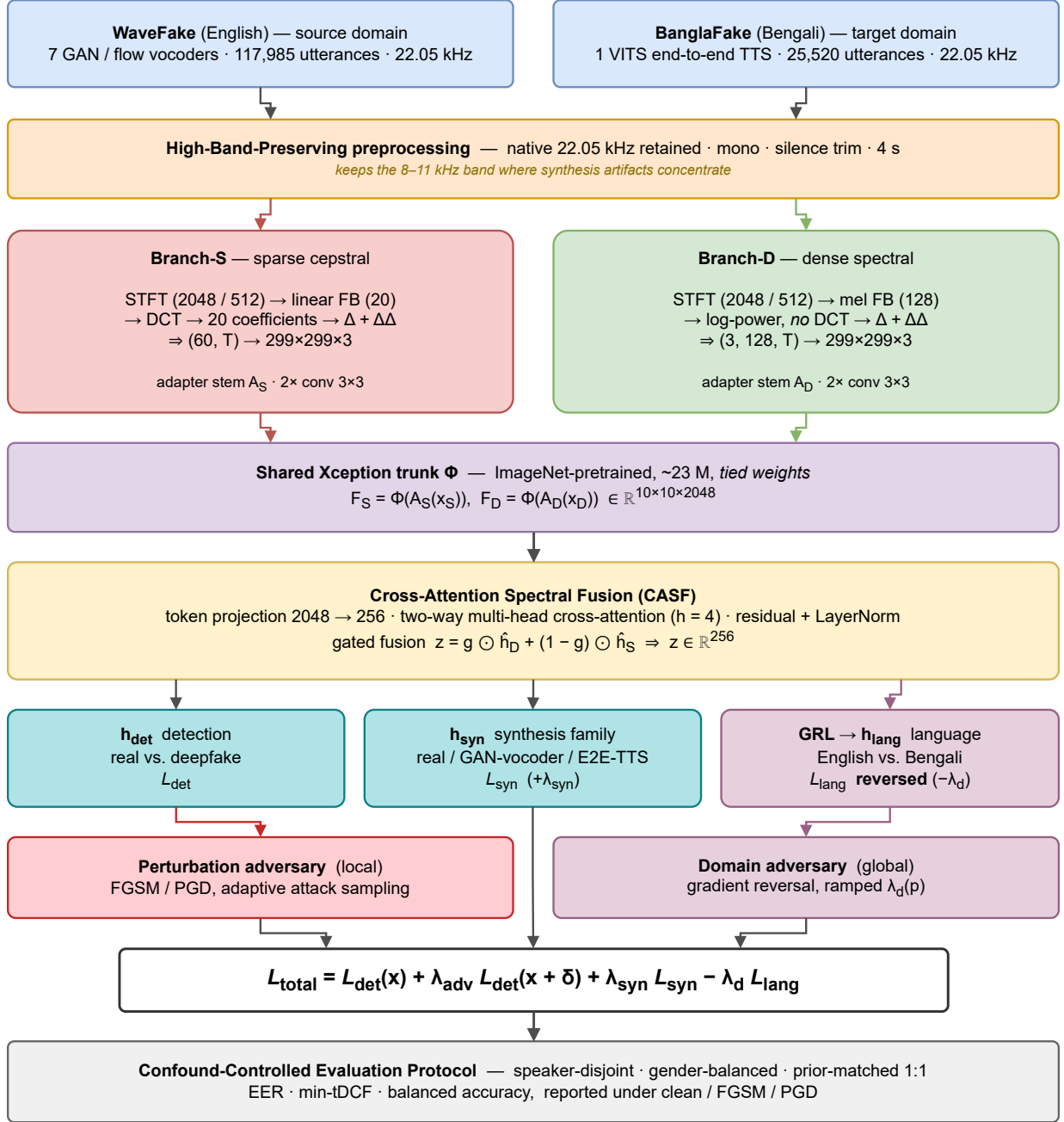


Fig. 1. End-to-end DASF-Net methodology. Both corpora pass through identical High-Band-Preserving preprocessing, split into a sparse cepstral and a dense spectral branch, and are encoded by one Siamese-shared Xception trunk. Cross-Attention Spectral Fusion produces a single embedding z on which three heads act: detection, an auxiliary synthesis-family classifier, and a gradient-reversed language discriminator. Two adversaries — perturbation and domain — shape the same embedding through the combined objective, and all reported figures follow the Confound-Controlled Evaluation Protocol.

any honest evaluation must neutralise, and that we address in Section VI:

- 1) **Speaker and gender.** BanglaFake’s synthetic side is one male speaker while its real side spans seven speakers of both genders. A detector can score highly by classifying gender rather than artifacts.
- 2) **Class prior.** WaveFake is roughly 8:1 synthetic-to-real;

BanglaFake is roughly 1:1. Accuracy is not comparable across such different priors, and a threshold calibrated on one is miscalibrated on the other.

- 3) **Bandwidth.** Both corpora are natively 22.05 kHz. Downsampling to 16 kHz, as in our prior study and much prior work, discards everything above 8 kHz — precisely the band in which vocoder and end-to-end TTS

artifacts are most pronounced, and the band most likely to differ between the two generator families.

Table I summarises both corpora after the alignment of Section IV-A.

TABLE I
DATASET SUMMARY AFTER HIGH-BAND-PRESERVING ALIGNMENT

	WaveFake	BanglaFake
Role	source	target
Language	English	Bengali
Real utterances	13,100	12,260
Synthetic utterances	104,885	13,260
Synthesis family	7, GAN / flow	1, VITS end-to-end
Speaker(s), synthetic	1 female	1 male
Speaker(s), real	1 female	7 (mixed gender)
Source corpus	LJSpeech	SUST TTS Corpus
Sample rate (retained)	22,050 Hz	22,050 Hz

IV. PROPOSED ARCHITECTURE: DASF-NET

Figure 1 gives the whole method; Fig. 2 expands the network itself. We describe each component and the reason it is present.

A. High-Band-Preserving preprocessing

Every utterance from both corpora is brought to a common representation: mono, native 22.05 kHz *retained*, silences longer than 2 s trimmed at a 40 dB threshold, and duration normalised to 4 s by centre-cropping or reflection padding. Retaining the native rate is a deliberate departure from. Down-sampling to 16 kHz imposes an 8 kHz Nyquist ceiling and removes the 8–11 kHz band; for a within-corpus study that loss is tolerable, but for a cross-*paradigm* claim it silently deletes the evidence most likely to distinguish a GAN vocoder from an end-to-end model. All filterbanks below therefore run to $f_{\max} = 11.025$ kHz.

B. Dual-branch spectral front-end

Both branches share Short-Time Fourier Transform parameters ($n_{\text{fft}} = 2048$, $\text{hop} = 512$), so any difference between them is attributable to the filterbank and compression stages alone.

Branch-S (sparse cepstral). Linear-frequency filterbank (20 filters, 0–11.025 kHz) $\rightarrow$ Discrete Cosine Transform $\rightarrow$ 20 cepstral coefficients $\rightarrow$ first and second temporal derivatives, giving a $(60, T)$ representation.

Branch-D (dense spectral). Mel-scaled filterbank ($n_{\text{mels}} = 128$, $f_{\max} = 11.025$ kHz) $\rightarrow$ log-power compression, with *no* DCT, so the dense time–frequency structure survives $\rightarrow$ first and second derivatives, giving a $(3, 128, T)$ tensor.

Both are resized to $299 \times 299 \times 3$, the Xception input geometry. Branch-S preserves the sharp, high-frequency periodicities on a linear scale that made LFCC effective for statistical back-ends; Branch-D preserves the dense spatial structure that convolution exploits. The two are complementary rather than competing, which is the premise of the fusion module.

C. Branch adapters and a Siamese-shared trunk

Each branch owns a small *adapter stem* A_S , A_D — two 3×3 convolutions with batch normalisation and ReLU, about 0.06 M parameters each — whose role is to map branch-specific input statistics into a common range before the shared encoder sees them.

The encoder Φ is a single ImageNet-pretrained Xception (≈ 23 M parameters) applied to both branches with tied weights:

$$F_S = \Phi(A_S(x_S)), \quad F_D = \Phi(A_D(x_D)), \quad (1)$$

with $F_S, F_D \in \mathbb{R}^{10 \times 10 \times 2048}$. Weight sharing is a design commitment, not an economy: two independent backbones would let each branch build a private feature space, and the fusion module would then have to reconcile two unrelated geometries. One trunk forces both views into a single artifact space at the cost of 0.12 M extra adapter parameters, and keeps the model comparable in size to the single-branch baseline of.

D. Cross-Attention Spectral Fusion (CASF)

Each feature map is flattened into 100 spatial tokens and linearly projected to $d = 256$, giving $T_S, T_D \in \mathbb{R}^{100 \times 256}$. Two multi-head cross-attention blocks ($h = 4$) let each view interrogate the other,

$$\hat{h}_D = \text{LN}(T_D + \text{MHA}(Q=T_D, K=T_S, V=T_S)), \quad (2)$$

$$\hat{h}_S = \text{LN}(T_S + \text{MHA}(Q=T_S, K=T_D, V=T_D)), \quad (3)$$

and a learned gate mixes them elementwise before mean-pooling over tokens:

$$g = \sigma(W_g [\hat{h}_D; \hat{h}_S] + b_g), \quad (4)$$

$$z = \text{pool}(g \odot \hat{h}_D + (1 - g) \odot \hat{h}_S) \in \mathbb{R}^{256}. \quad (5)$$

The gate is the point of the module. A fixed concatenation would assert that both views always matter equally; the gate instead lets the network decide, per utterance and per dimension, which representation carries the artifact. It also makes the model interpretable in a way the earlier feature comparison was not: the distribution of g over a test set is a direct, quantitative answer to “does this generator leave its trace in dense spectral structure or in sparse cepstral structure?”, and we report it as an analysis in its own right.

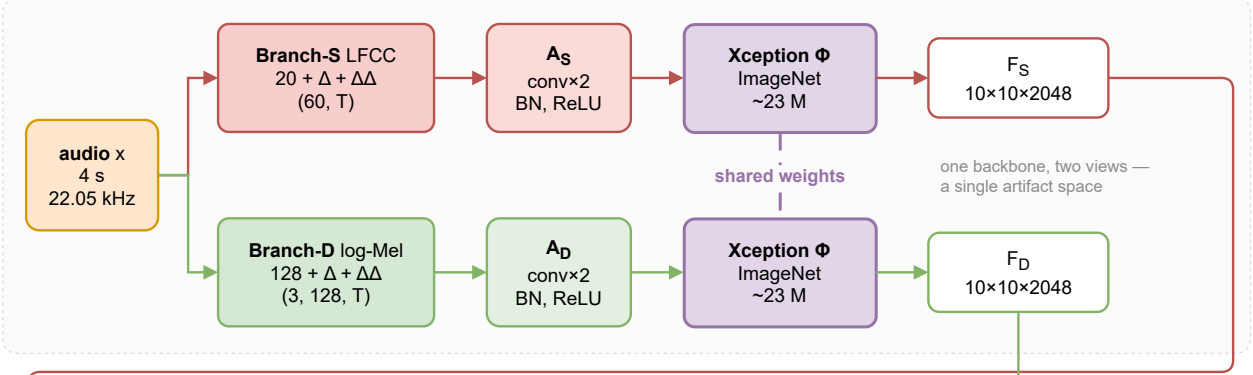
E. Objective heads

Three heads read the same z .

Detection head h_{det} ($256 \rightarrow 128 \rightarrow 2$) performs the task of record, real versus deepfake, under cross-entropy with label smoothing 0.1, giving $\mathcal{L}_{\text{det}}$.

Synthesis-family head h_{syn} ($256 \rightarrow 128 \rightarrow 3$) predicts $\{\text{real}, \text{GAN-vocoder}, \text{end-to-end TTS}\}$. This auxiliary supervision is deliberately coarse: predicting the specific generator would invite memorisation of generator identity, which is exactly the failure that breaks cross-vocoder transfer, whereas predicting the *family* pushes z to organise around mechanism. It contributes $\mathcal{L}_{\text{syn}}$ with a positive weight λ_{syn} .

(a) dual-branch encoder, Siamese weight sharing



(b) cross-attention spectral fusion and the three objective heads

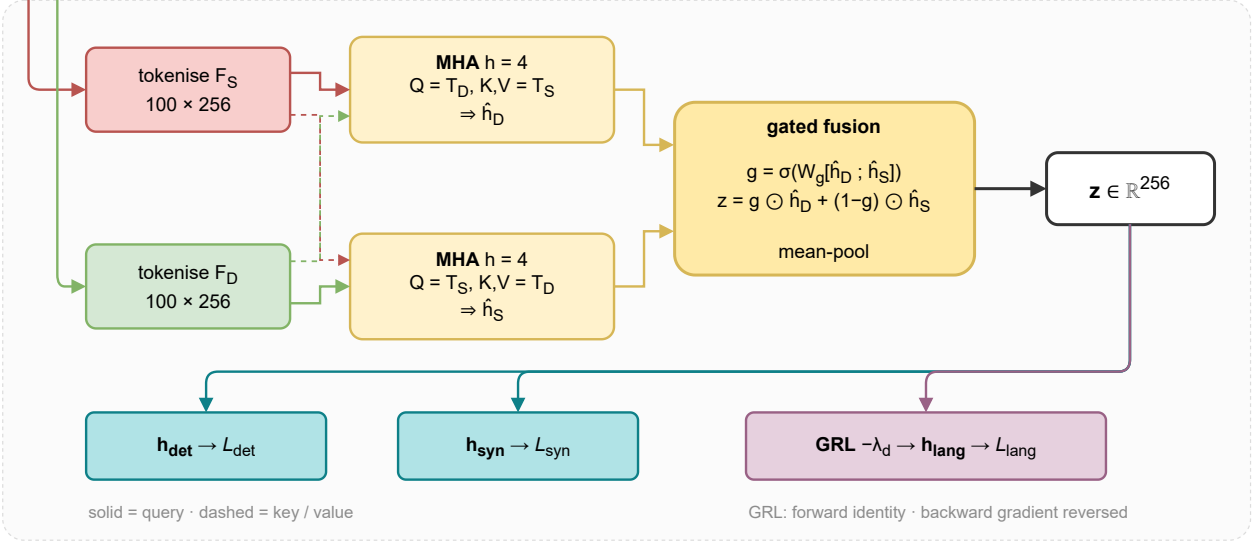


Fig. 2. DASF-Net. (a) The dual-branch encoder: one utterance becomes a sparse cepstral and a dense spectral view, each passed through its own adapter stem into a single weight-shared Xception trunk. (b) CASF tokenises both feature maps, lets each view attend over the other, and combines them through a learned gate into z ; three heads then act on z , the language head behind a gradient reversal layer.

Language head h_{lang} ($256 \rightarrow 128 \rightarrow 2$) predicts English versus Bengali, but sits behind a gradient reversal layer (GRL). The GRL is the identity in the forward pass and multiplies the gradient by $-\lambda_d$ in the backward pass. The head therefore does its best to identify the language while the shared trunk and CASF are driven to make that identification impossible. At convergence z retains what distinguishes synthetic from genuine and discards what distinguishes Bengali from English.

F. The dual-adversarial objective

Writing δ for an adversarial perturbation of the input features, the complete training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}(x) + \lambda_{\text{adv}} \mathcal{L}_{\text{det}}(x + \delta) + \lambda_{\text{syn}} \mathcal{L}_{\text{syn}} - \lambda_d \mathcal{L}_{\text{lang}}, \quad (6)$$

where the negative sign is realised by the GRL rather than by ascending the loss directly. Equation (6) contains two adversaries of different kinds. The perturbation adversary is

local: it searches a small neighbourhood of each input for a direction that flips the decision. The domain adversary is *global*: it searches the embedding for any direction that reveals the corpus. Our hypothesis is that they are complementary — that a representation stripped of language identity offers fewer spurious directions for a gradient attack to exploit, and that a representation hardened against local perturbation is less able to rely on the brittle, corpus-specific cues the domain adversary attacks. The ablation in Section VI is designed to test that hypothesis rather than assume it.

G. Threat model

We evaluate two white-box attacks in the normalised feature domain, where inputs lie in $[0, 1]$ after per-utterance standardisation. FGSM takes a single step,

$$\delta_{\text{FGSM}} = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{\text{det}}(x, y)), \quad (7)$$

and PGD iterates $K = 10$ steps with step size $\alpha = \epsilon/4$, projecting onto the ℓ_∞ ball of radius ϵ after each step. Critically, *both attacks are swept over the same ϵ grid*, $\epsilon \in \{0.001, 0.005, 0.010\}$. We note this explicitly because prior work, including our own earlier study, reports FGSM and PGD at ϵ values two orders of magnitude apart, which makes the resulting robustness figures non-comparable between the two attacks; results obtained under mismatched budgets should not be read as evidence that one attack is stronger than the other. All robustness numbers in this work are reported per ϵ .

V. TRAINING METHODOLOGY

A. Why a curriculum

Training Eq. (6) from random initialisation is unstable in practice: a gradient-reversed discriminator applied to an untrained encoder produces a large, uninformative reversed gradient, and adversarial examples generated against an untrained decision boundary are meaningless. We therefore introduce difficulty in three stages, illustrated in Fig. 3, each initialised from the previous.

Stage 1 — representation warm-up. Φ is frozen at its ImageNet weights; only A_S , A_D , CASF and h_{det} train, on labelled source data, under $\mathcal{L}_{\text{det}}$ alone ($\lambda_{\text{syn}} = \lambda_d = \lambda_{\text{adv}} = 0$). Adam, $lr = 10^{-4}$, batch 128, mixed precision. This establishes a usable fused representation before the pretrained trunk is disturbed.

Stage 2 — cross-lingual alignment. Φ unfreezes; h_{syn} and h_{lang} are attached. Batches mix labelled source audio with *unlabelled* target audio: target labels are never used, only the language tag, which keeps the setting honestly unsupervised with respect to the target domain. Following the reversal coefficient is ramped rather than fixed,

$$\lambda_d(p) = \lambda_{\text{max}} \left(\frac{2}{1 + e^{-10p}} - 1 \right), \quad (8)$$

with $p \in [0, 1]$ the fraction of the stage completed, so the discriminator gains influence only as it becomes informative.

Stage 3 — dual-adversarial hardening. All modules fine-tune at $lr = 5 \times 10^{-5}$ for 10 epochs with adversarial examples introduced by the adaptive sampler of Algorithm 2, with the language adversary still active. Attaching robustness after alignment, rather than before, means the perturbation adversary hardens a representation that is already language-invariant.

B. Adaptive attack sampling

Rather than fixing the ratio of FGSM to PGD batches, we sample the attack for each minibatch in proportion to a momentum-smoothed estimate of how successful each attack has recently been. Writing s_a for the running success estimate of attack a and $\hat{s}_a$ for its measured success on the current batch,

$$s_a \leftarrow (1 - m) s_a + m \hat{s}_a, \quad m = 0.2, \quad (9)$$

and attacks are drawn with probability proportional to s_a over the non-clean fraction, while a fixed fraction $p_c = 1/3$ of every

Algorithm 1 DASF-Net forward pass

Require: waveform w ; adapters A_S, A_D ; shared trunk Φ ; projection W_p ; attention blocks; gate W_g ; heads $h_{\text{det}}, h_{\text{syn}}, h_{\text{lang}}$; GRL coefficient λ_d

Ensure: detection logits $\hat{y}$, family logits $\hat{f}$, language logits $\hat{\ell}$, gate g

- 1: $w \leftarrow \text{HBP}(w)$ {22.05 kHz, trim, 4 s}
 - 2: $S \leftarrow \text{STFT}(w; n_{\text{fft}}=2048, \text{hop}=512)$
 - 3: $x_S \leftarrow \text{resize}(\Delta\Delta(\text{DCT}(\text{LinFB}_{20}(S))))$
 - 4: $x_D \leftarrow \text{resize}(\Delta\Delta(\log \text{MelFB}_{128}(S)))$
 - 5: $F_S \leftarrow \Phi(A_S(x_S)); F_D \leftarrow \Phi(A_D(x_D))$ {tied weights}
 - 6: $T_S \leftarrow W_p \text{tokens}(F_S); T_D \leftarrow W_p \text{tokens}(F_D)$
 - 7: $\hat{h}_D \leftarrow \text{LN}(T_D + \text{MHA}(T_D, T_S, T_S))$
 - 8: $\hat{h}_S \leftarrow \text{LN}(T_S + \text{MHA}(T_S, T_D, T_D))$
 - 9: $g \leftarrow \sigma(W_g[\hat{h}_D; \hat{h}_S] + b_g)$
 - 10: $z \leftarrow \text{pool}(g \odot \hat{h}_D + (1 - g) \odot \hat{h}_S)$
 - 11: $\hat{y} \leftarrow h_{\text{det}}(z); \hat{f} \leftarrow h_{\text{syn}}(z)$
 - 12: $\hat{\ell} \leftarrow h_{\text{lang}}(\text{GRL}_{\lambda_d}(z))$ {identity forward, $-\lambda_d$ backward}
 - 13: **return** $\hat{y}, \hat{f}, \hat{\ell}, g$
-

minibatch stays clean to preserve baseline accuracy. Training therefore concentrates on whichever attack currently works, without a hand-tuned schedule.

C. Algorithms

Algorithm 1 states the forward computation; Algorithm 2 states the training procedure.

D. Complexity

DASF-Net adds 0.12 M adapter parameters, a 2048×256 projection, two four-head attention blocks over 100 tokens, and three small MLP heads — together under 2 M parameters on top of the 23 M Xception trunk. Because Φ is shared, the dominant training cost is that the trunk runs twice per utterance, roughly doubling the forward and backward cost of the single-branch baseline; Stage 3 adds the usual adversarial-training overhead, with PGD at $K = 10$ dominating. Inference cost is two trunk passes plus attention over 100 tokens, which remains real-time on a single modern GPU.

VI. EXPERIMENTAL DESIGN

This section specifies the study to be run. No results are reported.

A. Research questions

- **RQ1 (representation).** Does the dense-over-sparse advantage observed for CNNs on English GAN-vocoder audio hold on Bengali VITS audio, and does learned fusion beat either view alone?
- **RQ2 (cross-lingual transfer).** How far does the dual-adversarial objective close the zero-shot gap between English and Bengali, in both directions, relative to a non-adversarial baseline?

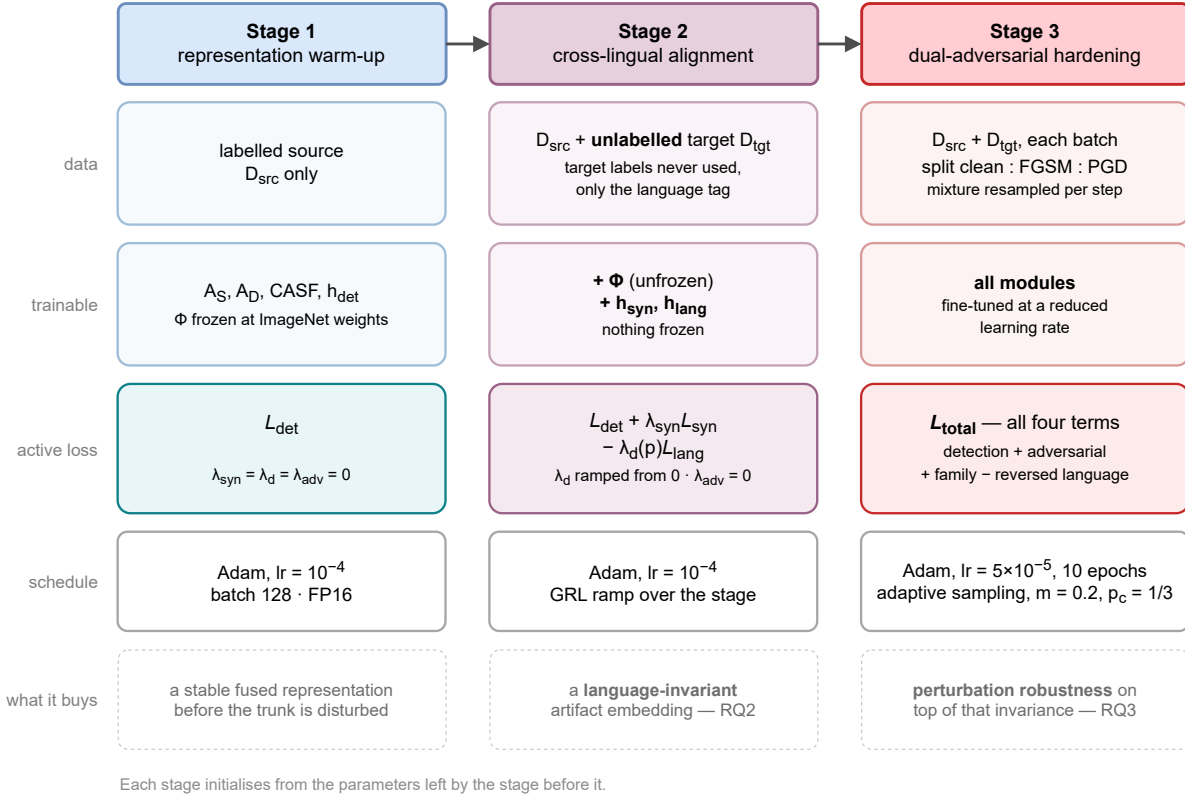


Fig. 3. The three-stage curriculum. Each stage adds one source of difficulty and inherits the weights of the stage before it: representation warm-up over labelled source data with the trunk frozen; cross-lingual alignment, which unfreezes the trunk and introduces the gradient-reversed language discriminator over unlabelled target audio; and dual-adversarial hardening, which adds adaptive FGSM/PGD sampling on top of the aligned representation.

- **RQ3 (robustness transfer).** Does adversarial robustness acquired on one language survive transfer to the other, and does language invariance make that transfer easier?

B. Confound-Controlled Evaluation Protocol

Every figure reported under this protocol is subject to four constraints, which follow directly from the confounds of Section III-C.

- 1) **Speaker-disjoint splits.** No speaker appears in both training and test partitions, so a model cannot succeed by speaker recognition.
- 2) **Gender-balanced Bengali evaluation.** Because BanglaFake pairs one male synthetic speaker against mixed-gender real speech, the primary Bengali test set is restricted to male real speech, and the mixed-gender result is reported separately as a diagnostic. A large gap between the two is itself evidence of gender shortcutting.
- 3) **Prior-matched test sets.** Both test sets are resampled to 1:1 real-to-synthetic, so that accuracy is interpretable and comparable across domains.
- 4) **Threshold-free primary metric.** EER is the headline metric, with min-tDCF and balanced accuracy along-

side it. Raw accuracy is never reported for a transfer condition without its matching prior.

C. Experiments

E1 — In-language baselines. Train and evaluate on each corpus separately: Branch-S only, Branch-D only, and full CASF fusion. Compare against the published ResNet18 fine-tuned baseline (79.17% accuracy, 0.2435 EER) and the single-branch Xception of our prior study. Addresses RQ1 and calibrates everything that follows.

E2 — Zero-shot transfer. Train on the source and evaluate on the target without target labels, in both directions: EN→BN, and BN→EN with a per-vocoder breakdown over the seven WaveFake generators. Each direction is run with and without the language adversary, which isolates the contribution of Stage 2. A few-shot variant fine-tunes on $\leq 10\%$ of target data to characterise how quickly the gap closes with minimal supervision. Addresses RQ2.

E3 — Robustness and its transfer. Under the threat model of Section IV-G, report clean, FGSM and PGD EER per ϵ for (i) a model adversarially trained and tested in-domain, and (ii) a model adversarially trained on the source and tested on the target. The gap between them measures how language-specific robustness is. Addresses RQ3.

Algorithm 2 Three-stage dual-adversarial training

Require: labelled source $\mathcal{D}_{\text{src}}$; unlabelled target $\mathcal{D}_{\text{tgt}}^u$; epoch budgets E_1, E_2, E_3 ; weights $\lambda_{\text{syn}}, \lambda_{\text{max}}, \lambda_{\text{adv}}$; momentum m ; clean fraction p_c ; budget grid $\mathcal{E}$

Ensure: trained parameters $\theta = \{A_S, A_D, \Phi, \text{CASF}, h_*\}$

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1: freeze  $\Phi$ ;  $s_{\text{FGSM}} \leftarrow s_{\text{PGD}} \leftarrow 0.5$ 
2: for  $e = 1$  to  $E_1$  do
3:   for all minibatch  $B \sim \mathcal{D}_{\text{src}}$  do
4:     update  $\{A_S, A_D, \text{CASF}, h_{\text{det}}\}$  on  $\mathcal{L}_{\text{det}}(B)$ 
5:   end for
6: end for
7: unfreeze  $\Phi$ ; attach  $h_{\text{syn}}, h_{\text{lang}}$ 
8: for  $e = 1$  to  $E_2$  do
9:   for all minibatch  $B = B_{\text{src}} \cup B_{\text{tgt}}, B_{\text{src}} \sim \mathcal{D}_{\text{src}}, B_{\text{tgt}} \sim \mathcal{D}_{\text{tgt}}^u$  do
10:     $p \leftarrow$  fraction of Stage 2 completed
11:     $\lambda_d \leftarrow \lambda_{\text{max}}(2/(1 + e^{-10p}) - 1)$ 
12:    update  $\theta$  on  $\mathcal{L}_{\text{det}} + \lambda_{\text{syn}}\mathcal{L}_{\text{syn}} - \lambda_d\mathcal{L}_{\text{lang}}$   $\{\mathcal{L}_{\text{det}}, \mathcal{L}_{\text{syn}}$ 
      on  $B_{\text{src}}$  only $\}$ 
13:   end for
14: end for
15: for  $e = 1$  to  $E_3$  do
16:   for all minibatch  $B$  do
17:     split  $B$  into  $B_{\text{clean}}$  ( $p_c$ ) and  $B_{\text{atk}}$  ( $1 - p_c$ )
18:      $a \sim \text{Categorical}(s_{\text{FGSM}}, s_{\text{PGD}})$  normalised
19:      $\epsilon \sim \text{Uniform}(\mathcal{E})$ 
20:      $\delta \leftarrow \text{ATTACK}_a(B_{\text{atk}}, \epsilon, \theta)$ 
21:      $\hat{s}_a \leftarrow$  misclassification rate of  $B_{\text{atk}} + \delta$ 
22:      $s_a \leftarrow (1 - m)s_a + m\hat{s}_a$ 
23:     update  $\theta$  on Eq. (6) over  $B_{\text{clean}} \cup (B_{\text{atk}} + \delta)$ 
24:   end for
25: end for
26: return  $\theta$ 
```

E4 — Ablation. Removing one component at a time from the full model: no CASF (concatenation instead of gated cross-attention); no shared trunk (independent backbones); no h_{syn} ; no GRL; no adversarial stage; and 16 kHz preprocessing in place of HBP. This is what converts an observed improvement into an attributed one, and the 16 kHz row in particular tests whether the bandwidth argument of Section IV-A holds empirically.

E5 — Gate analysis. Report the distribution of the fusion gate g per generator family. This asks directly whether GAN vocoders and end-to-end TTS leave their traces in different representations, and gives the fusion module an interpretable output rather than only an accuracy delta.

D. Implementation

PyTorch with mixed-precision training, Adam throughout, batch size 128, label smoothing 0.1. Adversarial attacks are implemented in the normalised feature domain described in Section IV-G. Every experiment is run with three random seeds and reported as mean and standard deviation; single-seed

results are not reported. Code, split definitions and evaluation scripts will be released with the empirical paper.

VII. LIMITATIONS AND THREATS TO VALIDITY

We state these in advance because several bound what the planned study can conclude, regardless of its outcome.

One Bengali generator. BanglaFake contains a single VITS system, so this design tests cross-*paradigm* generalisation but cannot test cross-vocoder generalisation *within* Bengali the way WaveFake’s seven-generator leave-one-out protocol does for English. A claim that DASF-Net generalises across Bengali synthesisers would not be supported by this data, and we do not intend to make one. Extending BanglaFake with a Bengali FastSpeech2 or Tacotron2 system is the natural next step.

Confounds mitigated, not eliminated. CCEP controls speaker, gender and prior, but each control shrinks the usable Bengali evaluation set, trading statistical power for validity. Both the controlled and uncontrolled figures should be read together.

White-box attacks only. FGSM and PGD assume full gradient access. Black-box, transfer, perceptual and physical-channel attacks are outside this design, and robustness claims should be read as bounded by that.

Two languages. Bengali and English differ in phoneme inventory, prosody and recording conditions simultaneously. Language invariance demonstrated over one pair is evidence, not proof, of invariance in general.

Adversarial cost. Stage 3 roughly doubles training time on top of the doubled forward cost of the two-branch encoder, which matters for reproduction on modest hardware.

VIII. CONCLUSION

We have presented DASF-Net, an architecture that treats the two standing weaknesses of audio deepfake detectors — brittleness across languages and brittleness under perturbation — as instances of the same problem, and attacks both with adversaries acting on one shared embedding. The architecture contributes a dual-branch spectral front-end over a Siamese-shared Xception trunk, a gated cross-attention fusion module that learns per utterance which spectral view carries the artifact, and a combined objective in which a gradient-reversed language discriminator and adaptive FGSM/PGD training shape the same representation. We have specified the three-stage curriculum that makes this trainable, the algorithms in full, a corrected and unit-consistent adversarial threat model, and a Confound-Controlled Evaluation Protocol that neutralises the speaker, gender and prior imbalances which otherwise make Bengali detection results unreadable.

What the design buys, if the hypothesis holds, is a detector whose features describe what a synthesiser did rather than which corpus it was trained on. Whether it does hold is an empirical question, and the experiments of Section VI — in particular the ablation that separates the fusion module from the language adversary from the perturbation adversary — are constructed to answer it in a way that attributes any gain to a specific cause. Those results are reported separately.